

**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION**

**MONITORING AND REPORTING PROGRAM 2014-0153-DWQ-R5332 FOR
PLUMAS EUREKA COMMUNITY SERVICES DISTRICT
AND
THREE TREES HOLDING II LLC
PLUMAS COUNTY**

This Monitoring and Reporting Program (MRP) describes requirements for monitoring a wastewater treatment system. This MRP is issued pursuant to Water Code section 13267. The Discharger shall not implement any changes to this MRP unless and until a revised MRP is issued by the Regional Water Quality Control Board (Regional Water Board) Executive Officer.

The State Water Resources Control Board (State Water Board) and Regional Water Boards are transitioning to the paperless office system. In some regions, Dischargers will be directed to submit reports (both technical and monitoring reports) to the State Water Board's Electronic Content Management (ECM) database via email in portable document format (pdf). The email address for the ECM submittal is centralvalleyredding@waterboards.ca.gov.

Water Code section 13267 states, in part:

“In conducting an investigation specified in subdivision (a), the regional board may require that any person who has discharged, discharges, or is suspected of having discharged or discharging, or who proposes to discharge waste within its region, or any citizen or domiciliary, or political agency or entity of this state who has discharged, discharges, or is suspected of having discharged or discharging, or who proposes to discharge, waste outside of its region that could affect the quality of waters within its region shall furnish, under penalty of perjury, technical or monitoring program reports which the regional board requires. The burden, including costs, of these reports shall bear a reasonable relationship to the need for the report and the benefits to be obtained from the reports. In requiring those reports, the regional board shall provide the person with a written explanation with regard to the need for the reports and shall identify the evidence that supports requiring that person to provide the reports.”

Water Code section 13268 states, in part:

“(a) Any person failing or refusing to furnish technical or monitoring program reports as required by subdivision (b) of section 13267, or failing or refusing to furnish a statement of compliance as required by subdivision (b) of section

13399.2, or falsifying any information provided therein, is guilty of a misdemeanor and may be liable civilly in accordance with subdivision (b).

(b)(1) Civil liability may be administratively imposed by a regional board in accordance with article 2.5 (commencing with section 13323) of chapter 5 for a violation of subdivision (a) in an amount which shall not exceed one thousand dollars (\$1,000) for each day in which the violation occurs.”

The Discharger owns and operates the wastewater system that is subject to the Notice of Applicability (NOA) of Water Quality Order 2014-0153-DWQ. The reports are necessary to ensure that the Discharger complies with the NOA and General Order. Pursuant to Water Code section 13267, the Discharger shall implement this MRP and shall submit the monitoring reports described herein.

All samples shall be representative of the volume and nature of the discharge or matrix of material sampled. The name of the sampler, sample type (grab or composite), time, date, location, bottle type, and any preservative used for each sample shall be recorded on the sample chain of custody form. The chain of custody form must also contain all custody information including date, time, and to whom samples were relinquished. If composite samples are collected, the basis for sampling (time or flow weighted) shall be approved by Regional Water Board staff.

Field test instruments (such as those used to test pH, dissolved oxygen, and electrical conductivity) may be used provided that they are used by a State Water Board California Environmental Laboratory Accreditation Program certified laboratory, or:

1. The user is trained in proper use and maintenance of the instruments;
2. The instruments are field calibrated prior to monitoring events at the frequency recommended by the manufacturer;
3. Instruments are serviced and/or calibrated by the manufacturer at the recommended frequency; and
4. Field calibration reports are maintained and available for at least three years.

AEROBIC TREATMENT UNIT MONITORING WWTP #6

Influent Monitoring

Influent samples shall be taken from a location that provides representative samples of the wastewater and flow rate. At a minimum, influent monitoring shall consist of the following:

Table 1. WWTP #6 Influent Monitoring Frequency

Constituent	Units	Sample Type	Sample Frequency	Reporting Frequency
Total Nitrogen	milligrams per liter (mg/L)	Grab	Monthly	Quarterly

Effluent Monitoring

Samples of effluent shall be taken at an area that is representative of the effluent quality distributed to the disposal area. At a minimum, effluent monitoring shall consist of the following:

Table 2. WWTP #6 Effluent Monitoring Frequency

Constituent	Units	Sample Type	Sample Frequency	Reporting Frequency
Flow Rate	gallons per day (gpd)	Metered (Flow rate may be metered or estimated based on potable water supply meter readings or another approved method. Flow rates may be measured as influent or effluent flow.)	Continuous	Quarterly
Biochemical Oxygen Demand	milligrams per liter (mg/L)	Grab	Monthly	Quarterly
Turbidity	nephelometric turbidity unit (NTU)	Grab	Daily	Quarterly
Settleable Matter	milligrams per liter (mg/L)	Grab	Monthly	Quarterly
Total Nitrogen	milligrams per liter (mg/L)	Grab	Monthly	Quarterly

Aerobic treatment units may be integrated in a treatment train and all components shall be inspected to verify operational status. Because aerobic treatment units generate more biosolids than septic systems (similar to the activated sludge process), systems shall be inspected and/or pumped at least as frequently as described below. Depending upon the amount of solids removed from the aerobic treatment unit, less frequent inspections may be allowed by the Regional Water Board's Executive Officer.

Inspections of sludge and scum depth are not required if the tanks are pumped at least annually.

Table 3. WWTP #6 Aerobic Treatment Unit Monitoring Frequency

Parameter	Units	Measurement Type	Inspection/Reporting Frequency
Sludge depth and scum thickness in each compartment of each tank	Feet	Staff Gauge	Quarterly
Distance between bottom of scum layer and bottom of outlet device	Inches	Staff Gauge	Quarterly
Distance between top of sludge layer and bottom of outlet device	Inches	Staff Gauge	Quarterly
Effluent filter condition (if equipped, clean as needed)	Not Applicable	Not Applicable	Quarterly

Aerobic treatment units shall be pumped when any one of the following conditions exist:

1. The combined thickness of sludge and scum exceeds one-third of the tank depth of the first compartment.
2. The scum layer is within 3 inches of the outlet device.
3. The sludge layer is within 8 inches of the outlet device.

All pumping reports shall be submitted with the next regularly scheduled monitoring report. At a minimum, the record shall include the date, nature of service, service company name, and service company license number.

ACTIVATED SLUDGE MONITORING WWTP #7

Influent Monitoring

Influent samples shall be taken from a location that provides representative samples of the wastewater and flow rate. At a minimum, influent monitoring shall consist of the following:

Table 4. WWTP #7 Influent Monitoring Frequency

Constituent	Units	Sample Type	Sample Frequency	Reporting Frequency
Total Nitrogen	milligrams per liter (mg/L)	Grab	Monthly	Quarterly

Effluent Monitoring

Samples of effluent shall be taken at an area that is representative of the effluent quality distributed to the disposal area. At a minimum, effluent monitoring shall consist of the following:

Table 5. WWTP #7 Effluent Monitoring Frequency

Constituent	Units	Sample Type	Sample Frequency	Reporting Frequency
Flow Rate	gallons per day (gpd)	Metered (Flow rate may be metered or estimated based on potable water supply meter readings or another approved method. Flow rates may be measured as influent or effluent flow.)	Continuous	Quarterly
Biochemical Oxygen Demand	milligrams per liter (mg/L)	Grab	Monthly	Quarterly
Total Suspended Solids	milligrams per liter (mg/L)	Grab	Monthly	Quarterly
Settleable Matter	milliliters per liter (ml/L)	Grab	Monthly	Quarterly
Total Nitrogen	milligrams per liter (mg/L)	Grab	Monthly	Quarterly

Constituent	Units	Sample Type	Sample Frequency	Reporting Frequency
Total Coliform Organisms	most probable number per 100 mL sample (MPN/100 ml)	Grab	Monthly	Quarterly

DISINFECTION SYSTEM MONITORING WWTP #6 and WWTP #7

When disinfection is performed, samples shall be collected from immediately downstream of the disinfection system. Depending upon the level of disinfection and wastewater disposal, monitoring requirements vary. Disinfection monitoring shall be customized to the site-specific conditions:

Table 6. WWTP #6 and #7 Disinfection System Monitoring Frequency

Constituent	Units	Sample Type	Sample Frequency	Reporting Frequency
Total Coliform Organisms	most probable number per 100 mL sample (MPN/100 ml)	Grab	Weekly	Quarterly
Turbidity	nephelometric turbidity unit (NTU)	Grab/Meter	Daily	Quarterly
Chlorine Residual	milligrams per liter (mg/L)	Grab	Daily	Quarterly

SUBSURFACE DISPOSAL AREAS

Dynamite Hill and WWTPs #6 and #7 backup leach fields

Subsurface disposal areas may be configured many different ways (e.g. traditional leach field, pressure-dosed, drip system, mound/at grade, gravel less, etc.). In general, monitoring shall be sufficient to determine if wastewater is evenly applied, the disposal area is not saturated, burrowing animals and/or deep-rooted plants are not present, and odors are not present. Inspection of dosing pump controllers, automatic distribution valves, etc. is required to maintain optimum treatment in the disposal area (and any sand or media filter if present). Monitoring shall include, at a minimum, the following:

Table 7. Subsurface Disposal Area Monitoring Frequency

Constituent	Inspection Frequency	Reporting Frequency
Pump Controllers, Automatic Valves, etc. (All pump controllers and automatic distribution valves shall be inspected for proper operation as recommended by the manufacturer.)	Quarterly	Quarterly
Nuisance Odor Condition	Quarterly	Quarterly
Saturated Soil Conditions (Inspect a disposal area for saturated conditions. If a mound system is used, inspect perimeter base for signs of wastewater seepage or saturated soil conditions.	Quarterly	Quarterly
Plant Growth (Shallow-rooted plants are generally desirable, deep-rooted plants such as trees shall be removed as necessary.)	Quarterly	Quarterly
Vectors or Animal Burrowing (Evidence of animals burrowing shall be immediately investigated, and burrowing animal populations controlled as necessary.)	Quarterly	Quarterly
Seepage Pit Condition (Seepage pits shall be inspected to ensure they are allowing wastewater to infiltrate as designed. Visual inspection of the water level in the seepage pit is adequate.)	Quarterly	Quarterly

RECYCLED WATER MONITORING

Golf Course Irrigation

If recycled water is used for irrigation of landscape areas, priority pollutant monitoring is required at the production facility. Sampling shall be consistent with the following:

Table 8. Recycled Water Monitoring Frequency

Constituent	Inspection Frequency	Reporting Frequency
Priority Pollutants	5 years	The next annual report

Landscape areas are defined as parks; greenbelts, playgrounds; school yards; athletic fields; golf courses; cemeteries; residential landscaping; common areas; commercial landscaping (except eating areas); industrial landscaping (except eating areas); freeway, highway, and street landscaping.

LAND APPLICATION AREA MONITORING

Golf Course Irrigation

The Discharger shall monitor LAAs when wastewater and/or supplemental irrigation water is applied. If wastewater/supplemental irrigation water is not applied during a reporting period, the monitoring report shall so state. The Discharger shall monitor LAA as specified in the following table:

Table 9. Land Application Area Monitoring Frequency

Constituent	Units	Sample Type	Sampling Frequency	Reporting Frequency
Supplemental Irrigation	gallons per day (gpd)	Meter (see a below)	Monthly	Quarterly
Wastewater Flow	gallons per day (gpd)	Meter (see a below)	Monthly	Quarterly
Local Rainfall	Inches	Weather Station (see b below)	Monthly	Quarterly
Acreage Applied (see c below)	Acres	Calculated	Monthly	Quarterly
Application Rate (see d below)	gallon/acre/month	Calculated	Monthly	Quarterly
Soil Erosion Evidence	Not Applicable	observation	Monthly	Quarterly
Containment Berm Condition	Not Applicable	observation	Monthly	Quarterly
Soil Saturation/Ponding	Not Applicable	observation	Monthly	Quarterly
Nuisance Odors/Vectors	Not Applicable	observation	Monthly	Quarterly
Discharge Off-Site	Not Applicable	observation	Monthly	Quarterly

Table Notes:

- Meter requires meter reading, a pump run time meter, or other approved method.
- Weather station may be site-specific station or nearby governmental weather reporting station.
- Acreage applied denotes the acreage to which wastewater is applied.
- Application rate may also be reported as inch/acre/month. Table 7. Land Application Area Monitoring Frequency.

SOLIDS DISPOSAL MONITORING

WWTPs #6 and WWTP #7

The Discharger shall report the handling and disposal of all solids (e.g., screenings, grit, sludge, biosolids, etc.) generated at the wastewater systems. Records shall include the name/contact information for the hauling company, the type and amount of waste transported, the date removed from the wastewater systems, the disposal facility name and address, and copies of analytical data required by the entity accepting the waste. These records shall be submitted as part of the annual monitoring report.

GROUNDWATER MONITORING

The Discharger shall monitor groundwater quality as required by the NOA and this MRP. The discharger shall monitor the groundwater monitoring network which consists of three groundwater monitoring wells (GW-1 through GW-3) and five french drains (FD-1 through FD-5, formerly known as GW-4 through GW-8)

Consistent with the Business and Professions Code, groundwater monitoring reports, well construction workplans, etc. shall be prepared under the supervision of a California licensed civil engineer or geologist. Prior to construction of any groundwater monitoring wells the Discharger shall submit plans and specifications to the Regional Water Board's staff for review and approval. Once installed, all monitoring wells designated as part of the monitoring network shall be sampled and analyzed according to the schedule below.

The data from routine groundwater monitoring events shall be submitted quarterly. Analysis of the data and groundwater flow directions shall be performed at least annually and shall be performed under the supervision of a California licensed professional (as described above). The Discharger may request a reduced monitoring and reporting schedule once adequate data has been collected to characterize the site. (Typically two years of quarterly sampling is required for adequate characterization.

Prior to sampling, groundwater elevations shall be measured, and the wells shall be purged of at least three well volumes and until pH and electrical conductivity have stabilized. No-purge, low-flow, or other sampling techniques are acceptable if they are described in an approved Sampling and Analysis Plan. Depth to groundwater shall be measured to the nearest 0.01 feet. Groundwater elevations shall be calculated. Samples shall be collected using approved USEPA methods. Groundwater monitoring shall be monitored as specified in the following table:

Table 10. Groundwater Monitoring Frequency

Constituent	Units	Sample Type	Sampling/Reporting Frequency (see d below)
Groundwater Elevation (see a below)	0.01 Feet	Calculated	Quarterly

Constituent	Units	Sample Type	Sampling/Reporting Frequency (see d below)
Depth to Groundwater	0.01 Feet	Measurement	Quarterly
Gradient	Feet/Foot	Calculated	Quarterly
pH	Std. Units	Calculated	Quarterly
Electrical Conductivity	micromhos per centimeter (Umhos/cm)	Grab	Quarterly
Total Dissolved Solids	milligrams per liter (mg/L)	Grab	Quarterly
Nitrate as Nitrogen	milligrams per liter (mg/L)	Grab	Quarterly
Chloride	milligrams per liter (mg/L)	Grab	Quarterly
Total Coliform Organisms (see b below)	most probable number per 100 mL sample (MPN/100 ml)	Grab	Quarterly

Table Notes:

- a. Groundwater elevation shall be based on depth to water using a surveyed measuring point elevation on the well and a surveyed reference elevation.
- b. Using a minimum of 15 tubes or three dilutions.
- d. Analysis of data by a California licensed professional is required at least annually.

SURFACE WATER MONITORING

The Discharger shall monitor surface water quality as required by the NOA and this MRP. The discharger shall monitor surface water at the following seven locations as summarized in following table:

Sample ID	Sampling Location Description
RW-1	Middle Fork of the Feather River at Mohawk Road Bridge
RW-2	Intermittent watercourse adjacent to the 16 th tee area of the golf course
RW-3	Madora Creek along the north side of Unit 7B prior to entering the Middle Fork of the Feather River
RW-4	Madora Creek at the intersection of the Middle Fork of the Feather River, just before mixing with the river
RW-5	Middle Fork of the Feather River north of the Plumas Pines Golf Course's #4 green

Sample ID	Sampling Location Description
RW-6	Middle Fork of the Feather River north of the Plumas Pines Golf Course's #2 green
RW-7	Middle Fork of the Feather River downstream of Plumas Pines Country Club

Because of the difficulty in monitoring bacteria in surface water, sample collection procedures must be described in a Sampling and Analysis Plan. Natural bacteria levels can vary significantly, and may be correlated with rainfall. When possible, surface water bacteria samples should be collected under dry weather conditions. It is critical when monitoring bacteria that all containers and surfaces that a sample contacts are sterile. Sample containers must be autoclaved or manufactured to maintain sterility; use of screw top bottles, Whirl-pak® bags, or similar containers are acceptable. The sample hold time for bacteria samples is typically no more than six hours. Monitoring shall include, at a minimum, the following:

Table 11. Surface Water Monitoring Frequency

Constituent	Units	Weather (Rain/Dry)	Sampling Frequency	Reporting Frequency
Escherichia coli (E. coli) (see a below)	most probable number per 100 mL sample (MPN/100 ml)	Observation	Quarterly	Quarterly
Enterococci (see b below)	most probable number per 100 mL sample (MPN/100 ml)	Observation	Quarterly	Quarterly

Table Notes:

- a. Analysis by USEPA Method 1603 or equivalent.
- b. Analysis by USEPA Method 1600 or equivalent.

REPORTING

In reporting monitoring data, the Discharger shall arrange the data in tabular form so that the date, sample type (e.g., effluent, solids, etc.), and reported analytical or visual inspection results are readily discernible. The data shall be summarized to clearly illustrate compliance with the General Order and NOA as applicable. The results of any monitoring done more frequently than required at the locations specified in the MRP

shall be reported in the next regularly scheduled monitoring report and shall be included in calculations as appropriate.

During the life of this General Order, the State Water Board or Regional Water Board may require the Discharger to electronically submit monitoring reports using the State Water Board's California Integrated Water Quality System (CIWQS) program Internet web site or alternative database. Electronic submittal procedures will be provided when directed to begin electronic submittals. Until directed to electronically submit monitoring reports, the Discharger shall submit hard copy monitoring reports.

A. Quarterly Monitoring Reports

Quarterly reports shall be submitted to the Regional Water Board on the **first day of the second month after the quarter ends** (e.g. the January-March Quarterly Report is due by May 1st). Monthly monitoring shall be submitted in Quarterly Reports. The reports shall bear the certification and signature of the Discharger's authorized representative. At a minimum, the quarterly reports shall include:

1. Results of all required monitoring.
2. A comparison of monitoring data to the discharge specifications, applicable effluent limits, disclosure of any violations of the NOA and/or General Order, and an explanation of any violation of those requirements. (Data shall be presented in tabular format.)
3. If requested by staff, copies of laboratory analytical report(s) and chain of custody form(s).

B. Annual Report

Annual Reports shall be submitted to the Regional Water Board **with the 4th Quarter Monitoring Report**. The Annual Report shall include the following:

1. Tabular and graphical summaries of all monitoring data collected during the year.
2. An evaluation of the performance of the wastewater treatment facility, including discussion of capacity issues, nuisance conditions, system problems, and a forecast of the flows anticipated in the next year. A flow rate evaluation as described in the General Order (Provision E.2.c) shall also be submitted.
3. A discussion of compliance and the corrective action taken, as well as any planned or proposed actions needed to bring the discharge into compliance with the NOA and/or General Order.
4. A discussion of any data gaps and potential deficiencies/redundancies in the monitoring system or reporting program.

5. The name and contact information for the wastewater operator responsible for operation, maintenance, and system monitoring.
6. A groundwater monitoring report prepared by a California licensed professional. This report may be prepared separately from the rest of the Annual Report. The report shall contain an analysis of groundwater data collected during the year. The analysis shall include a description of the sample events, copies of the field logs, purge method and volume, groundwater elevation and trend, a groundwater elevation map for each sample event, summary tables showing results for parameters measured, comparison of groundwater quality parameters to standards in the NOA, chain-of-custody forms, calibration logs for field equipment used, and a general evaluation of any impacts the wastewater discharge is having on groundwater quality.

A letter transmitting the monitoring reports shall accompany each report. The letter shall report violations found during the reporting period, and actions taken or planned to correct the violations and prevent future violations. The transmittal letter shall contain the following penalty of perjury statement and shall be signed by the Discharger or the Discharger's authorized agent:

"I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of the those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment."

The Discharger shall implement the above monitoring program as of the date of this MRP.

Ordered by:

Original signed by Bryan J. Smith, P.E. for
PATRICK PULUPA, Executive Officer

2/12/2020

(Date)